

Equations and Inequalities Problem Solving

Lesson 7-7

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



'Twice a number is 18' $\rightarrow 2n = 18$ — the equation models the words

Model

Write the definition:

$$x + 2 = 7$$

balanced with =

$x + 5 = 12$ — exactly one answer: $x = 7$

Equation

Write the definition:

$$x + 5 > 12 \text{ — many answers: } x = 8, 9, 10, \dots$$

Inequality

Write the definition:

247

hundreds tens ones

If the problem asks for a number of people and you get $x = -3$, that is NOT reasonable

Reasonableness

Write the definition:



stands for a number

*"5 less than a number is 12" becomes $x - 5 = 12$,
where x is the variable.*

Variable

Write the definition:



number

"You can spend at most \$20" is a constraint: $\text{cost} \leq 20$.

Constraint

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can model and solve real-world problems using equations and inequalities.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Model

Equation

Inequality

Reasonableness

Variable

Constraint

- 1 A drawing, equation, or diagram that represents a situation is a _____.
- 2 A math sentence with an equal sign is an _____.
- 3 A math sentence that compares values with $<$, $>$, $\leq$, or $\geq$ is an _____.
- 4 Checking whether an answer makes sense for the problem is checking its _____.
- 5 A letter that stands for the unknown amount in a problem is a _____.
- 6 A limit that tells which values are allowed in a problem is a _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: A museum has some paintings. After adding 12, they have 45. Which equation models this?

A. $p + 12 = 45$

B. $p - 12 = 45$

C. $12p = 45$

D. $p / 12 = 45$

Step 1 'Adding 12' to the original number gives 45: $p + 12 = 45$.

Step 2 Solve: $p = 33$.

 **Answer:** A. $p + 12 = 45$

 We do — together

Solve this one with your class using the same steps.

Problem: A detective needs more than 8 hours to finish the investigation. She has already worked 3 hours. Which inequality represents the additional hours h she needs?

A. $3 + h > 8$

B. $3 + h = 8$

C. $h > 3$

D. $8 + h > 3$

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Which model is correct for: 'A number divided by 6 equals 7'?

A. $n / 6 = 7$

B. $6n = 7$

C. $n - 6 = 7$

D. $n + 6 = 7$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Room 3: A sign warns, 'The safe holds at most 20 gold bars.' Which inequality models the number of bars x ?

- A. $x \leq 20$
- B. $x \geq 20$
- C. $x > 20$
- D. $x = 20$

Show your work:

2. Room 4: A locked drawer clue reads, 'Five identical evidence boxes weigh 60 pounds in all.' Solve $5w = 60$ for each box's weight w .

- A. $w = 12$
- B. $w = 55$
- C. $w = 300$
- D. $w = 65$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Create two word problems about the same situation: one that requires an equation and one that requires an inequality. Solve both and explain why the models are different.

Sentence starter: Equation problem: _____. Model: _____. Solution: _____. Inequality problem: _____. Model: _____. Solution: _____. The models are different because _____.

Show your work:

Reflect — Exit Ticket

A box of donuts has some donuts. After giving away 7, there are fewer than 5 left. Which inequality represents the starting number of donuts d ?

- A. $d - 7 < 5$
- B. $d + 7 < 5$
- C. $d - 7 > 5$
- D. $d - 7 = 5$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Model
2. **Notes 2:** Equation
3. **Notes 3:** Inequality
4. **Notes 4:** Reasonableness
5. **Notes 5:** Variable
6. **Notes 6:** Constraint
7. **We Try (notes):** A. $3 + h > 8$ — *3 hours plus additional hours h must be more than 8: $3 + h > 8$. Solve: $h > 5$.*
8. **You Try (notes):** A. $n / 6 = 7$ — *Divided by 6 is $n / 6$. Equals 7 means $= 7$. So $n / 6 = 7$.*
9. **Try It 1:** A. $x \leq 20$ — *'At most 20' means 20 or fewer, so $x \leq 20$.*
10. **Try It 2:** A. $w = 12$ — *Divide both sides by 5: $w = 60 \div 5 = 12$ pounds. Check: $5 \times 12 = 60$ ✓*
11. **Exit Ticket:** A. $d - 7 < 5$ — *Starting with d donuts and giving away 7 leaves fewer than 5: $d - 7 < 5$. Solve: $d < 12$. ✓*